Exercise-1

OBJECTIVE QUESTIONS

A-1. In insulators, the forbidden energy gap is very large, in case of semiconductor it is moderate and in conductors it is very small or zero.

D-35. ac ⇨ Rectifier ⇨ dc

D-37. OC is reverse bias while potential of point A is approx 0.7 volt.

Hence answer is (3).

F-9. The output D for the given combination

D = $\overline{(A+B).C} = \overline{(A+B)} + \overline{C}$

If A = B = C = 0 then D = $\overline{(0+0)} + \overline{0} = \overline{0} + \overline{0}$ = 1 + 1 = 1

If A = B = 1, C = 0 then D = $\overline{(1+1)} + \overline{0} = \overline{1} + \overline{0}$ = 0 + 1 = 1

Exercise-2

PART - I

1. By using E = $\frac{V}{d} = \frac{0.6}{10^{-6}}$ = 6×10^5 V/m

2. The diode is in reverse bias so current through it is zero.

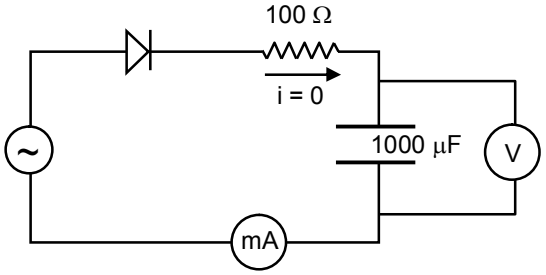
3. Current flow is possible and $i = \frac{V}{R} = \frac{(4-1)}{300}$ = 10^{-2} A

4. For full wave rectifier $\eta = \frac{81.2}{1+\frac{r_f}{R_L}}$ $\Rightarrow$ n_{max} = 81.2% ($r_f << R_L$)

10. α is the ratio of collector current and emitter current while β is the ratio of collector current and base current.

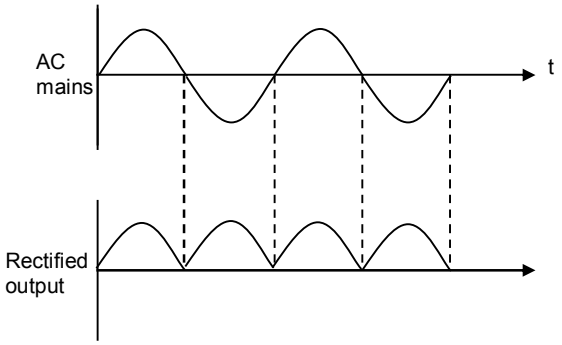
11. $\beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98}$ = 49

18.



In this state the circuit will work as half wave rectifier and the voltage across capacitor will be equal to average value of voltage of ac source over a half cycle.

19. frequency doubles = $100H_2$



21. (1,3,4) based on practical information.

Exercise-3

PART - II

2. Increase of temperature increases resistance of conductor but decreases resistance of semiconductor.

5. Cu is conductor and Ge is semiconductor.
Increase of temperature increases resistance of conductor but decreases resistance of semiconductor.

8. When NPN transistor is used as an amplifier, majority charge carrier electrons of N-type emitter move from emitter to base and than base to collector.

12. $I = n_e A v_d$

$$\frac{I_e}{I_h} = \frac{n_e \times (v_d)_e}{n_h \times (v_d)_h}$$

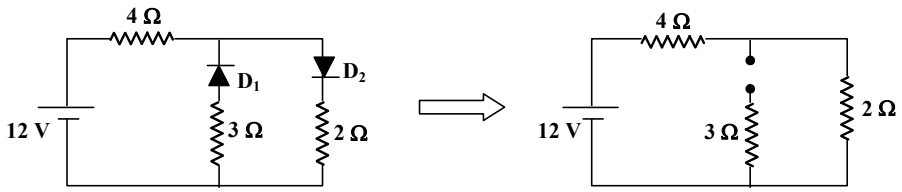
Here, $\frac{n_e}{n_h} = \frac{7}{5}$, $\frac{I_e}{I_h} = \frac{7}{4}$

$$\frac{7}{4} = \frac{7}{5} \times \frac{(v_d)_e}{(v_d)_h} \Rightarrow \frac{(v_d)_e}{(v_d)_h} = \frac{5}{7} \times \frac{7}{4} = \frac{5}{4}.$$

13. $\beta = \frac{I_C}{I_B}$ and $I_E = I_C + I_B$ $\therefore$ $\beta = \frac{I_C}{I_E - I_C} = \frac{5.488}{5.60 - 5.488} = 49.$

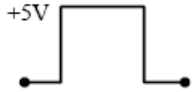
14. If lattice constant of semiconductor is decreased, then E_C and E_V increase, but E_g decreases.

15.



amp $\therefore$ (4) is correct. सही है।

16. Diode is forward biased in first half cycle and amplitude of signal is 5V.



Correct choice: (4)

20. **Ans. (4)**

$Y = (\overline{\bar{A} + \bar{B}}) = A.B$

it is AND gate.

21. **Ans. (2)**

From Half wave rectifier

22. $(\overline{A + B})$ = NOR gate

When both inputs of NAND gate are connected, it behaves as NOT gate.

$\Rightarrow$ OR + NOT = NOR.

$\Rightarrow$ OR + NOT = NOR

23. $y = \overline{\left(\overline{A \cdot \overline{A \cdot B}}\right) \cdot \left(\overline{B \cdot \overline{A \cdot B}}\right)}$

$= \left(\overline{A \cdot \overline{A \cdot B}}\right) \cdot \left(\overline{B \cdot \overline{A \cdot B}}\right)$

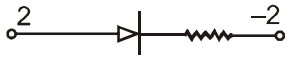
$= A \cdot \left(\bar{A} + \bar{B}\right) + B \cdot \left(\bar{A} + \bar{B}\right)$

$= A \cdot \bar{A} + A \cdot \bar{B} + B \cdot \bar{A} + B \cdot \bar{B}$

$y = 0 + A \cdot \bar{B} + B \cdot \bar{A} + 0$

24. For same value of current higher value of voltage is required for higher frequency hance (1) should be correct answers.

25. For forward bias $V_P > V_N$

So 

is forward biased.

26. For conductor (Cu) resistance increases linearly and for semiconductor resistance decreases Exponentially in given temperature range.

27. whenever we have 1 at input, output is 1.

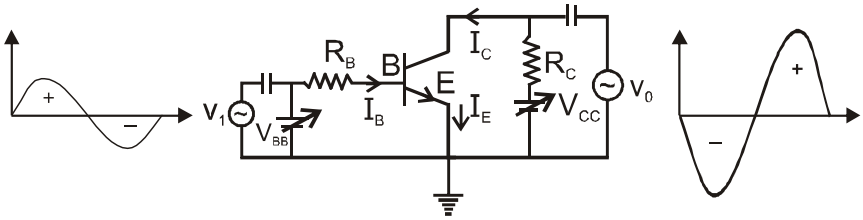
so the gate is or

28. From standard data
Ans. is (4)

29. $\frac{1}{\alpha} = \frac{1}{\beta} + 1$

$\alpha = \frac{\beta}{\beta + 1}$

30.



$\Delta V_{CC} = \Delta V_{CE} + R_L \Delta I_C = 0$

or $\Delta V_{CE} = - R_L \Delta I_C$

The change in V_{CE} is the output voltage v_0. From equation we get

$v_0 = \Delta V_{CE} = - \beta_{ac} R_L \Delta I_B$

The voltage gain of the amplifier is

$$A_v = \frac{v_0}{v_i} = \frac{\Delta V_{CE}}{r \Delta I_B} = - \frac{\beta_{ac} R_L}{r}$$

The negative sign represents that output voltage is opposite with phase with the input voltage.

31. For silicon diode barrior potential is 0.7V

so $I = \dfrac{3 - 0.7}{200}$

= 0.0115A

= 11.5mA